Control Unit – Steering Turret Interface Control Document (Is.1)

1. Command and Control Protocol between Control Unit and Steering Turret

1.1. Purpose

This section defines the communication protocol used between the Steering Turret (ST) and the Control Unit (CU) for the transmission and reception of operational commands, status requests, and acknowledgements.
The protocol provides a deterministic, versioned, and extensible binary interface ensuring reliable and low-latency communication suitable for real-time control applications.

1.2. Physical and Transport Layer

Parameter	Value
Medium	Ethernet
Transport Protocol	TCP/IP
Port Number (Steering Turret)	Configurable (default 7000)
Direction	Commands: CU → ST Responses (ACK/NACK): ST → CU
Transmission Rate	Event-driven (on request)
Encoding	Binary, Little-Endian (least significant byte first)
Character Set	Binary (no ASCII encoding)

TABLE 1 – COMMANDS PHYSICAL AND TRANSPORT LAYER

1.3. Command Message Format

Field	Size (bytes)	Type	Description
magic	4	uint32_t	Constant “CMD”
protocol_version	1	uint8_t	Identifies the protocol version. Current value: 0x01.
controller_id	1	uint8_t	Identifies the controller_ID
cmd_id	2	uint16_t	Unique command identifier.
seq	2	uint16_t	Sequence number for matching the response to the command. Wraps at 65535.
timestamp	8	uint64_t	Epoch timestamp
payload_size	2	uint16_t	Size in bytes of the payload following the header.
payload	variable	binary	Command-specific arguments.

controller_id	***Description***
11	Control Unit
13	Logger application.

15	Diagnostic application
21	Simulator or other test applications

1.4. Command Definitions

cmd_id	*Description*	*Payload*	*Response Payload*
0x0001	CU_STATE	uint8_t cu_state	—
0x0002	SELFLEV	uint8_t start_stop	—
0x0003	IBIT	uint8_t start_stop	—
0x0004	STORAGE	—	—
0x0005	LASER	uint8_t id, uint8_t on_off	—
0x0006	TURRET	uint8_t m_type, float az, float el	—
0x0008	SUT	uint8_t id, uint8_t m_type, float az, float el	
0x000A	RESET	—	—
0x000C	FACTORY	—	—
0x000E	SHUTDOWN	—	—
0x000F	CONFIG		—
0x0010	MANUALPOS	float lat, float, long, float alt, float heading	—
0x0011	POSMODE	uint8_t mode	—
0x0012	REQ_SW_VERSION	—	string ASCII string (e.g. "CU_SW_v1.0")
0x0013	REQ_HW_INFO	—	ASCII string or structured HW info
0x0014	REQ_WORK_TIME	—	uint32_t uptime_sec

TABLE 2 – COMMANDS LIST

1.4.1. CU_STATE - Set State

Size of payload: 1 byte

Field	Size (bytes)	Type	Description
cu_state	1	uint8_t	State_ID

state_id	*state*	*Description*
0	INIT	
1	INIT DONE	
2	SAFE	
3	OPER	
4	SELF_LEVELLING	Performing Self-Levelling
5	IBIT_RUNNING	IBIT is running
6	STORE	Motors in storage position
0xFF	FAIL	

1.4.2. SELFLEV - Start/Stop Self-level

Size of payload: 1 byte

Field	Size (bytes)	Type	Description
StartStop	1	uint8_t	1=Start, 0=Stop

1.4.3. IBIT - Start/Stop IBIT

Field	Size (bytes)	Type	Description
StartStop	1	uint8_t	1=Start, 0=Stop

1.4.4. LASER - Set Laser ON/Off

Field	Size (bytes)	Type	Description
id	1	uint8_t	Laser ID (1-8)
OnOff	1	uint8_t	Switch On/Off the selected laser

1.4.5. Turret – Modify the azimuth of the turret and the elevation of the camera

Field	Size (bytes)	Type	Description
Motion_type	1	uint8_t	Motion Type (0=relative, 1 absolute, 2=speed)
az_data	4	float	azimuth value. • delta Deg for relative motion • absolute Deg for absolute pos • deg/s for speed
el_data	4	float	elevation value. • delta Deg for relative motion • absolute Deg for absolute pos • deg/s for speed

1.4.6. MIRROR – Modify the elevation one or all mirrors

Field	Size (bytes)	Type	Description
Mirror_ID	1	uint8_t	Identifier of the mirror (0xFF=all mirrors)
Motion_type	1	uint8_t	Type (0=relative, 1 absolute, 2=speed)
elevation_data	4	float	elevation value. • delta Deg for relative motion • absolute Deg for absolute pos • deg/s for speed

1.4.7. SUT – Modify the azimuth and mirror elevation of one or all SUTs

Field	Size (bytes)	Type	Description
SUT_ID	1	uint8_t	Identifier of the SUT (0xFF=all SUTs)
Motion_type	1	uint8_t	Type (0=relative, 1 absolute, 2=speed)
az_data	4	float	azimuth value. • delta Deg for relative motion • absolute Deg for absolute pos • deg/s for speed
el_data	4	float	elevation value. • delta Deg for relative motion • absolute Deg for absolute pos deg/s for speed

1.4.8. MANUALPOS – Turret position/heading used in manual position mode

Field	Size (bytes)	Type	Description
Lat	8	double	Degrees
Lon	8	double	Degrees
Alt	8	double	m
Heading	8	double	Degrees

1.4.9. POSMODE – Position mode Manual

Field	Size (bytes)	Type	Description
mode	1	uint8_t	Position mode (0=GPS, 1=Manual)

1.5. Response Message Format (ACK/NACK)

Response to command

Field	Size (bytes)	Type	Description
magic	4	uint32_t	Constant
protocol_version	1	uint8_t	Identifies the protocol version. Current value: 0x01.
cmd_id	2	uint16_t	Echo of the received command ID.
seq	2	uint16_t	Same sequence number as the originating command.
result	1	uint8_t	Result code: 0x00 = ACK, 0x01 = NACK.
err_code	2	uint16_t	Detailed error or reason code (0 = no error).
payload_size	2	uint16_t	Size in bytes of the additional response data.
payload	variable	binary	Optional data (e.g., return values).

err_code	*Description*
00	No error
01	Client unknown
02	Protocol mismatch
03	Syntax error
04	Value out of range
05	Not allowed in this state
06	Insufficient resources

For all commands except REPORTREQ, the payload of the response is empty

1.6. Sequence Handling and Error Management

- Each command shall include a sequence number incremented by the CU.
- The ST shall include the same sequence number in the response, ensuring command-response correlation.
- If the ST detects an invalid command ID, malformed packet, or parameter error, it shall reply with NACK and a corresponding err_code.

2. High Frequency Telemetry Protocol (ST → CU)

2.1. Purpose

This section defines the Telemetry Data Protocol used by the Steering Turret (ST) to transmit real-time high frequency positioning data to the Control Unit (CU).
The protocol is binary, deterministic, version-controlled, and suitable for real-time UDP transmission at 100 Hz.

2.2. Physical and Transport Layer

Parameter	Value
Medium	Ethernet
Transport Protocol	UDP/IP Multicast
Multicast IP	239.255.10.2
Port Number	Configurable (default 6010)
Direction	ST→ CU
Transmission Rate	100 Hz (every 10 ms)
Encoding	Binary, Little-Endian
Character Set	Not applicable
Reliability	Best-effort (UDP)
Timeout / Loss Handling	CM shall detect discontinuity in sequence numbers

TABLE 3 – TELEMETRY PHYSICAL AND TRANSPORT LAYER

2.3. High Frequency Telemetry Payload Format

Field	Type	Size (bytes)	Description
magic	uint32_t	4	Fixed identifier "TLMH". Used to recognize telemetry packets and verify packet integrity.
protocol_version	uint8_t	1	Identifies the telemetry format version. Current value = 1.
seq	uint16_t	2	Sequential counter incremented for each telemetry packet. Wraps at 65535.
timestamp	uint64_t	8	UNIX time in milliseconds.
position[18]	float[18]	72	Turret and SUT angular positions: [az_tur, el_tur, az_sut1, el_sut1, …].
rates[18]	float[18]	72	Turret and SUTangular rates [az_tur_rate, el_tur_rate, az_sut1_rate, el_sut1_rate, …].

TABLE 4 – TELEMETRY PAYLOAD FORMAT

3. Low Frequency Telemetry Protocol

3.1. Purpose

This section defines the Telemetry Data Protocol used by the Steering Turret (ST) to transmit real-time low frequency data to the Control Unit (CU).
The protocol is binary, deterministic, version-controlled, and suitable for real-time UDP transmission at 100 Hz.

3.2. Physical and Transport Layer

Parameter	Value
Medium	Ethernet
Transport Protocol	UDP/IP Multicast
Multicast IP	239.255.10.3
Port Number	Configurable (default 6020)
Direction	ST→ CU
Transmission Rate	100 Hz (every 10 ms)
Encoding	Binary, Little-Endian
Character Set	Not applicable
Reliability	Best-effort (UDP)
Timeout / Loss Handling	CM shall detect discontinuity in sequence numbers

TABLE 5 – LOW FREQUENCY TELEMETRY PHYSICAL AND TRANSPORT LAYER

3.3. Low Frequency Telemetry Format

This section describes the low-rate telemetry message (100 ms periodicity) transmitted from the Steering Turret (ST) toward the Control Unit (CU).
The low-rate telemetry provides health, environmental, state and diagnostic information.

Field	Type	Size	Description
magic	uint32_t	4	Fixed identifier "TLML" used to recognize the packet type and perform a first-level integrity check.
protocol_version	uint8_t	1	Telemetry protocol format version. Current valid value: 1.
controller_id	uint8_t	1	Identifier of the originating controller inside the ST subsystem.
seq	uint16_t	2	Monotonic sequence counter incremented for each packet. Wraps at 65535.
timestamp	uint64_t	8	UNIX time in milliseconds.
State	uint8_t	1	Current operational state of the Steering Turret.
Uptime	uint32_t	4	Time in seconds since the last system boot.
Temperature	float	4	Environmental internal temperature (°C).
Humidity	float	4	Environmental relative humidity (%).
Pressure	float	4	Atmospheric pressure (hPa).
Position Lat	double	8	Turret latitude (degrees).
Position Lon	double	8	Turret longitude (degrees).
Position Alt	double	8	Turret altitude (meters)
Turret Heading	double	8	Turret heading (degrees).
Motor status	uint8_t[8]	8	Status flags for each motorized axis/subassembly. Each byte represents one motor.
Laser status	uint8_t[8]	8	Status flags for laser alignment devices. Each byte represents one laser unit.
Level X	float	4	Platform inclination along the X-axis (degrees).

Level Y	float	4	Platform inclination along the Y-axis (degrees).
NumOfErrors	uint8_t	1	Number of error/warning codes included in the message (0–255).
Errorcodes	uint32_t*	variable	List of error and warning identifiers, encoded as described below.

3.4. Error and Warning Codes

The field Errorcodes conveys a compact diagnostic summary of the last detected system anomalies. It contains up to 256 entries, each encoded as a 32-bit unsigned integer.

The list is ordered, always with:
Errors first (blocking or critical anomalies)
Warnings next (non-blocking degradations or low-confidence conditions)

3.4.1. Structure of Each Error/Warning Code (32-bit)

Each diagnostic identifier is a 32-bit value with the following structure:

Byte 3	Byte 2	Byte 1	Byte 0
Category	Subcategory	Error Code	

Byte 3 – Category (0–255)
High-level functional area that generated the diagnostic entry.

Byte 2 – Subcategory (0–255)
Specific subsystem or component inside the functional area.

Bytes 1–0 – Error/Warning Code (0–65535)
Unique identifier of the anomaly condition.

3.4.2. Category / Subcategory Definition

Category	Meaning	Subcategories (examples)
0	General system	0 General subsystem
1	Pointing motors	0 Azimuth motor
		1 Elevation motor (Aiming Camera)
2	Mirror elevation motors	0 Mirror motor 1
		1 Mirror motor 2
		…
3	SUT azimuth motors	0 SUT motor 1
		1 SUT motor 2
		…
4	GPS / Orientation	0 GPS 1
		…
5	Self-Leveling	0 Self-Leveling actuator
		…
6	Environmental sensors	0 Sensor 1
		…
...	Extendable	Additional subsystems may be added as needed

3.4.3. Meaning of Errors vs Warnings

Errors
Indicate a critical malfunction or violation of operational constraints.
Examples: motor over-current, encoder failure, communication timeout, self-leveling failure.

Warnings
Non-critical impairments that may degrade performance but do not require leaving the Operational state.
Examples:
• GPS not fixed
• Low GPS accuracy
• Environmental sensor out of tolerance
• Motor temperature approaching limit

Warnings are typically associated with temporary degraded conditions.